
Unauthored by Design: The Seed’s Share of a Generated Artifact

Wenhao Lu
NXT Horizon
wenhao@nxthorizon.org

Adithya Balakumar
NXT Horizon
adithya.balakumar@nxthorizon.org

Hyunjun Yi
Seoul International School
Seoul, Republic of Korea
hjyi2027@gmail.com

Abstract

When an artist prompts a generative model, how much of the resulting artifact did they actually determine? We isolate the contribution of the random seed (the part of the output that exists before any human intent expressed in the prompt has a chance to matter) with a fully crossed prompt $\times$ model $\times$ seed design run identically over image and text generation, so the two modalities can be compared on the same footing. Across 240 images (two SDXL-class models, a six-rung “specificity ladder” of nested prompts from 3 to 35 words, 10 seeds per cell) and 240 text generations, we decompose output variance into orthogonal prompt, model, interaction, and seed components. Three findings bear on creative agency. First, the prompt dominates semantic variance in images (62.4% vs. 30.9% for the seed), but escalating a prompt from three words to thirty-five removes only about 69% of the seed’s semantic randomness: roughly 31% survives the most constrained specification we wrote, and the bootstrap band sits well clear of zero. Specification has a floor. Second, the seed is not pure noise: the same seed value carries a small but statistically consistent signature across unrelated prompts ($p = 0.001$ in both models), concentrated in spatial luminance structure: the seed behaves less like dice and more like an anonymous collaborator with a faint style. Third, in text the pattern inverts: added constraints halve *lexical* spread while *semantic* spread stays essentially where three words left it, and on the serverless provider we tested, the seed parameter is silently ignored, making the randomness not even addressable. We discuss what an irreducible, partly structured stochastic contribution means for where prompting ends and authorship begins.

1 Introduction

The public conversation about generative AI and authorship is mostly conducted over the prompt: whether writing one is a creative act, whether it is enough of one, and who owns what it yields. Far less attention goes to the other input the artist supplies, usually without noticing: the random seed. Every generation call resolves a seed before the first denoising step or the first sampled token; the seed fixes the initial noise tensor or the sampling trajectory, and everything the prompt does, it does *to* that randomness. If a large share of the finished artifact is determined by the seed alone, then a correspondingly large share of the work was authored by no one, or by something the artist can select but not intend.

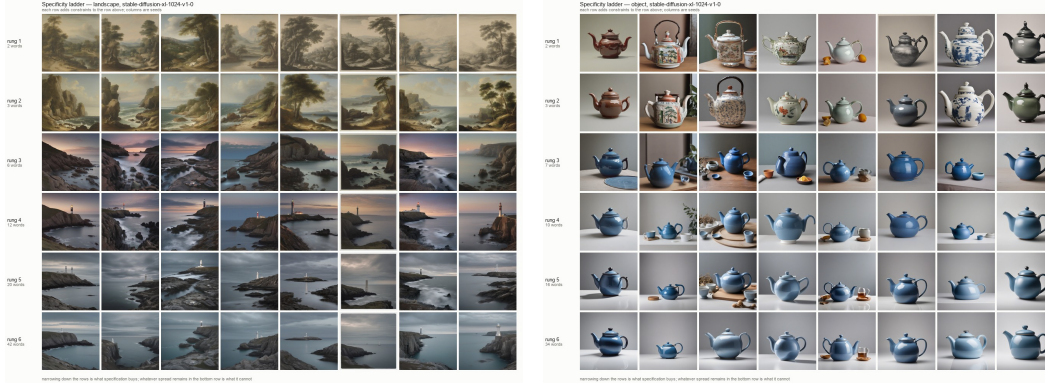


Figure 1: **The specificity ladder** (SDXL 1.0; left: landscape family, right: object family). Rows are nested prompt rungs from 3 words (top) to 35 words (bottom); columns are seeds, with all other generation parameters held constant. The narrowing down the rows is what specification buys. The spread that remains in the bottom row (differences in rock structure, cloud texture, teapot form) is what it cannot: about 31% of the semantic seed variance at rung 1 survives at rung 6.

This paper asks that question quantitatively, and asks it the same way for images and for text. The design is deliberately simple: fix the prompt, sweep the seed, and decompose the variance of the outputs. Because the seed is the only thing that varies within a (prompt, model) cell, within-cell variance *is* seed variance: no model internals, no attribution heuristics, just a balanced ANOVA over feature representations of the artifacts.

Three refinements turn the decomposition into an argument about agency rather than a table of percentages:

The specificity ladder. Any hand-picked prompt set makes the seed/prompt ratio a property of the picker. We instead use six *nested* rungs per prompt family, from "a landscape" (3 words) to a 35-word specification pinning down palette, light, composition, and exclusions. The question stops being "is the seed bigger than the prompt?" (which is unanswerable without a prompt distribution) and becomes "how much randomness can added specification actually remove, and does it ever reach zero?", which needs no prompt distribution at all. The residual at the top rung is the part of the artifact that more prompting cannot buy back.

The seed-signature test. The decomposition treats seeds as interchangeable replicates. But the same seed list runs against every prompt, so the assumption is testable: does seed k push unrelated prompts in a consistent direction? If yes, the seed is not noise but a faint, unchosen style, which changes what "selecting a seed" means as a creative act.

Intent legibility. Variance shares say how much the output moved; they do not say whether what the prompt asked for survived into the thing a viewer sees. We ask directly: given one artifact, can a classifier tell which prompt produced it? Intent that cannot be recovered from the work arguably was not exercised on it, the sharpest form of the agency question this study can answer without human subjects.

2 Related work

Variance decomposition of generative outputs connects to classical random-effects ANOVA (Searle et al., 1992), here applied to vector-valued artifacts by summing sums of squares over feature dimensions (equivalently, the trace of the Gower-centred cross-product matrix, as in distance-based MANOVA (Anderson, 2001)). The role of initial noise in diffusion models is studied from the optimisation side, where "golden noise" selection and noise-space search treat the seed as a latent worth optimising (Qi et al., 2024; Zhou et al., 2024), whereas we treat it as a source of unauthored variation to be measured. Work on prompt engineering and controllability (Liu and Chilton, 2022; Oppenlaender, 2023) implicitly assumes specification converges on intent; the specificity floor quantifies where that convergence stops. Within Creative AI discourse, distributed and negotiated agency between artists

and generative systems is a recurring critical theme (McCormack et al., 2019; Hertzmann, 2018); our contribution is an empirical instrument for one piece of that negotiation.

3 Method

Design. A fully crossed prompt $\times$ model grid with n seeds per cell, one generation per (prompt, model, seed) triple. Prompts form two families (*landscape*, *object*) $\times$ six nested rungs. Images: 12 prompts $\times$ 2 models (stable-diffusion-xl-1024-v1-0, playground-v2-1024px-aesthetic) $\times$ 10 seeds = 240 images, balanced across 24 cells, all sampler settings constant (1024 $\times$ 1024, 30 steps, CFG 7). Text: the same 12 rung-paired prompts $\times$ 1 model (gpt-oss-120b) $\times$ 20 replicates = 240 generations at temperature 1.0.

Feature views. Each artifact is represented in two views per modality. Images: `image_lowlevel`, 56 model-free perceptual statistics (palette, luminance distribution, spatial structure); and `image_semantic`, a VLM caption of the image embedded into 768 dimensions (Nussbaum et al., 2024). Text: `text_lexical` (surface statistics) and `text_semantic` (the same 768-d embedder). Agreement between a semantic and a model-free surface view guards against results being artifacts of the embedder.

Variance decomposition. With the design balanced and fully crossed, the decomposition

$$SS_{\text{total}} = SS_{\text{prompt}} + SS_{\text{model}} + SS_{\text{prompt} \times \text{model}} + SS_{\text{within}}$$

is exactly orthogonal, and SS_{within} is seed variance, since the seed is the only thing varying inside a cell. Seed is deliberately not a main effect: seed 7 under prompt A has no a priori relation to seed 7 under prompt B, so it enters as the replicate stratum. We report EMS-corrected variance components with prompt and model as random factors (we care what a *typical* prompt induces, not about these twelve), since raw η^2 inflates prompt and model shares with seed noise. Uncertainty comes from a two-level bootstrap (resampling prompts and seeds); prompt and model effects get permutation tests.

Specificity floor. Within each family, within-cell (seed) variance is tracked across rungs and expressed as a fraction of the 3-word rung’s. The value retained at rung 6 is the floor: randomness that the most detailed specification we wrote did not remove.

Seed signature. Prompt and seed form a complete crossed grid with one artifact per cell, the classic randomised-block layout: $F = MS_{\text{seed}}/MS_{\text{residual}}$, with significance from permuting seed labels independently within each prompt, which destroys cross-prompt consistency while leaving each prompt’s own spread untouched.

Intent legibility. A leave-one-out nearest-centroid classifier predicts the prompt from the artifact’s features, computed within each generative model so model identity cannot leak in as a cue. Accuracy above the 1/12 chance rate is the share of the specification readable off the work.

The pipeline is validated against synthetic data with planted variance splits, planted seed signatures, and planted floor plateaus; all analyses re-run from committed feature matrices with no API calls.

4 Results: images

The prompt wins the semantic layer; the seed holds the surface. In the semantic view the prompt accounts for 62.4% of variance against the seed’s 30.9%, with a seed/prompt ratio of 0.50 whose interval excludes 1. In the low-level view the ordering reverses (ratio 1.37), though the interval straddles 1, so seed dominance of the surface is suggested rather than established. Read together: *what* the image depicts is mostly the artist’s; *how* it is textured, lit, and spatially organised is substantially the seed’s.

Specification has a floor. Moving from rung 1 to rung 6 removes about 69% of semantic seed variance, and then stops. The retained 30.9% has a bootstrap band (0.123–0.199 in absolute variance) well clear of zero. Figure 1 shows what the number looks like: at the 35-word rung, every image is

Table 1: Image variance decomposition (240 images, EMS-corrected components). CI: 95% two-level bootstrap.

	image_lowlevel	image_semantic
variance from seed	35.5%	30.9%
variance from prompt	25.9%	62.4%
seed/prompt ratio (CI)	1.37 (0.84–2.19)	0.50 (0.31–0.67)
seed variance retained at rung 6	55.9%	30.9%
prompt recovered from artifact (chance 8.3%)	67.9%	74.6%
seed signature	$p = 0.001$, both models	$p \leq 0.002$, both models

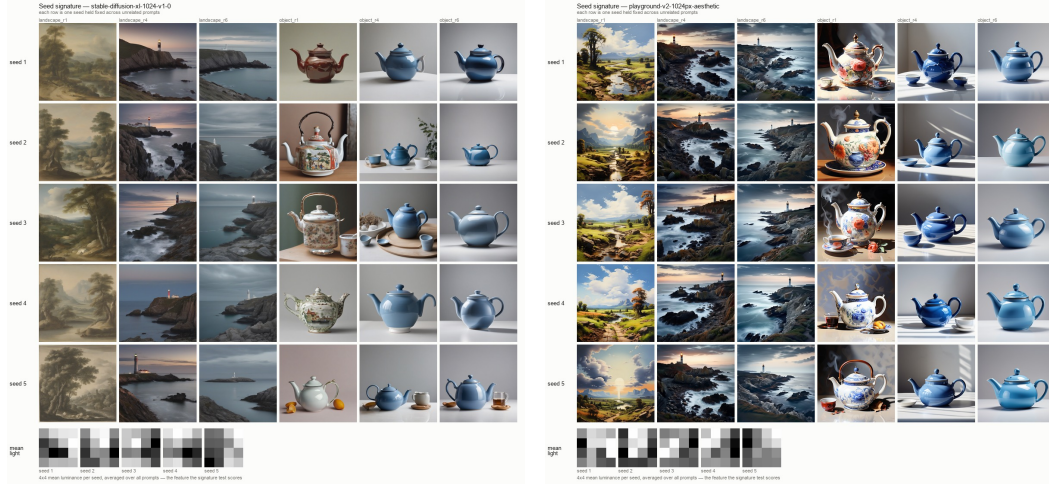


Figure 2: **The seed signature** (left: SDXL 1.0, right: Playground v2). Each row holds one seed fixed across unrelated prompts; the strip beneath shows per-seed mean 4×4 luminance maps. The images mostly show the prompt winning, which is what the statistics say too, but the luminance strips carry the signature: the same seed distributes light consistently across prompts ($p = 0.001$).

recognisably the specified scene, and no two are the same picture. Two honest wrinkles: the low-level curve *rises* at rungs 2–3 before falling (plausibly, "a landscape" collapses onto a narrow mode and "coastal . . . at dusk" opens up palette and lighting range before later constraints close it down; a hypothesis, not a result), and the decline is not monotone in either view (rung 5 exceeds rung 4), which 10 seeds per cell is too few to resolve.

The seed has a faint style. The signature test rejects the pure-noise hypothesis in both models ($p = 0.001$ low-level, $p \leq 0.002$ semantic), with the effect concentrated in spatial luminance features: a given seed tends to place light and dark in consistent regions across unrelated prompts (Figure 2). The effect is small next to the prompt’s, but its *stability across every prompt* is what makes it a style rather than a perturbation, and what makes “seed shopping” a selection among faint, unchosen collaborators rather than a lottery.

Intent is legible. The prompt is recoverable from a single artifact 74.6% of the time semantically (67.9% from model-free surface statistics alone) against 8.3% chance. Most of what the artist specified survives into the visible work.

5 Results: text

The paired text ladder ran on gpt-oss-120b: sampling variance is 21.8% (lexical) and 23.7% (semantic) against the prompt’s 78.2% / 76.3%, ratios 0.28 (CI 0.17–0.36) and 0.31 (0.23–0.38). Intent legibility is near-ceiling semantically (99.6%) and 72.5% lexically.

The floor is where text diverges from images. Across six rungs of tightening constraint, ending in a 60-word specification of exactly three declarative sentences, *lexical* within-cell variance roughly halves (100% $\rightarrow$ 50.7%), but *semantic* variance ends at 86.6% of where three words left it, never dipping meaningfully. The constraints regularise length and sentence shape while the semantic spread stays put: in this setting, prompting a language model is a purchase of form, much more than of content. One model, so we state this as a hypothesis.

Three caveats change how these numbers read. First, the column is *sampling* variance, not seed variance: the serverless provider we used (Fireworks) silently ignores the seed parameter on chat completions. We verified this across ten models, where a fixed seed at temperature 1.0 diverges within ~ 40 characters on repeat calls while temperature 0 is deterministic. Within-cell variance is real and is what we measure, but no text generation here is reproducible, and the seed-signature test is skipped as vacuous rather than reported as null. Second, nine of ten candidate text models leaked chain-of-thought into output or returned empty content at the tightest rungs; the surviving single model means the model and interaction terms read 0.0% by construction. Third, for these reasons the image and text seed shares are not measuring the same thing and should not be read as one cross-modal number: diffusion randomness is addressable and reproducible; this text randomness is neither.

That last asymmetry is itself a finding about agency. For images, the artist can at least *select* their randomness: revisit a seed, curate among faint styles, reproduce a result. For text on this class of provider, the stochastic collaborator is anonymous *and* unaddressable: a third of the semantic outcome (and, at the top rung, most of what constraints failed to pin) is contributed by a process no one can name, summon again, or sign.

6 Discussion: what the artist does with this

The study puts numbers on a boundary that debates about AI authorship usually gesture at. Roughly a third of an image’s semantic content, and more of its surface, is fixed before the prompt’s intent applies, and no amount of specification we could write drove that share to zero. This reframes several familiar practices. *Prompt escalation* is real but saturating: the first constraints buy the most, and past the fourth rung the artist is mostly regularising, not determining. *Seed shopping*, regenerating until something clicks, is not sampling from noise but curating among collaborators with faint, stable dispositions; the creative act it most resembles is casting, not commissioning. And the *legibility* results cut in the artist’s favour: what was specified is overwhelmingly recoverable from the work, so the unauthored share coexists with, rather than erases, the authored one.

For the track’s theme: agency here is neither possessed nor lost but *partitioned*, and the partition is measurable. The seed’s share is agency that belongs to no one: not the artist, who cannot intend it; not the model authors, who did not choose it; not the system, which merely resolves it. Creative practice with these tools is, in a precise sense, practice at working alongside an irreducible remainder.

7 Limitations

Two prompt families, hand-written ladders, and (for images) 10 seeds per cell: the intervals are wide, the non-monotonicities unresolved, and the floor value is a property of ladders like ours, not a universal constant. The semantic view routes through a captioner and an embedder; the model-free surface view agreeing with it is a guard, not a proof. The text results rest on one model and on provider behaviour that may change without notice. Legibility uses a classifier, not human viewers; the human-subjects version of the question remains open.

References

- M. J. Anderson. A new method for non-parametric multivariate analysis of variance. *Austral Ecology*, 26(1):32–46, 2001.
- A. Hertzmann. Can computers create art? *Arts*, 7(2):18, 2018.
- V. Liu and L. B. Chilton. Design guidelines for prompt engineering text-to-image generative models. In *Proc. CHI*, 2022.

- J. McCormack, T. Gifford, and P. Hutchings. Autonomy, authenticity, authorship and intention in computer generated art. In *EvoMUSART*, 2019.
- Z. Nussbaum, J. X. Morris, B. Duderstadt, and A. Mulyar. Nomic Embed: Training a reproducible long context text embedder. *arXiv:2402.01613*, 2024.
- J. Oppenlaender. A taxonomy of prompt modifiers for text-to-image generation. *Behaviour & Information Technology*, 2023.
- Z. Qi et al. Not all noises are created equally: Diffusion noise selection and optimization. *arXiv:2407.14041*, 2024.
- S. R. Searle, G. Casella, and C. E. McCulloch. *Variance Components*. Wiley, 1992.
- Z. Zhou et al. Golden noise for diffusion models: A learning framework. *arXiv:2411.09502*, 2024.